

Omkar Sudhir Patil

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Professional Summary

AI/ML researcher and control systems engineer specializing in intelligent autonomous systems. Pioneered Lyapunov-based Deep Neural Networks (LbDNNs), developing the first DNN controller with real-time adaptation and provable stability guarantees. Proven track record securing \$2M+ in funding, publishing 20+ papers (300+ citations), and deploying control solutions on real robotic platforms. Expert in bridging theoretical AI with practical engineering applications across aerospace, robotics, and autonomous systems.

Professional Experience

Research Scientist

Oct 2025 – Present

University of Florida, Department of Mechanical & Aerospace Engineering

- Leading research on next-generation mathematical frameworks for modular and compositional control design, enabling scalable solutions for complex autonomous systems
- Developing energy-based control methods leveraging geometric structures for improved stability and efficiency
- Advancing adaptive control solutions for aerospace, spacecraft servicing, and multi-agent robotics applications

Postdoctoral Research Associate

May 2023 – Oct 2025

University of Florida, Department of Mechanical & Aerospace Engineering

- Secured \$2M+ in competitive research funding from ONR, AFOSR, and AFRL for AI and control systems projects
- Pioneered Lyapunov-based deep neural network (Lb-DNN) controllers with online weight adaptation, enabling real-time learning without pre-training—solving a 25-year open problem in adaptive control
- Developed and deployed adaptive control algorithms for quadrotors, spacecraft, and multi-agent systems with mathematical stability guarantees, validated on hardware platforms
- Published 20 papers in top-tier venues (IEEE TAC, IEEE T-NN, ACC, CDC) with 300+ citations
- Mentored 15 graduate students and postdocs; created open-source MATLAB/Python packages deployed on robotic platforms

Graduate Research Assistant

Aug 2019 – May 2023

University of Florida, Nonlinear Controls and Robotics Group

- Solved 17-year open problem on exponential stability of RISE controllers—a fundamental result in robust control theory
- Developed composite adaptive learning algorithms using concurrent learning, achieving parameter convergence without restrictive excitation conditions
- Implemented and validated controllers on quadrotors, robotic manipulators, and multi-agent testbeds with real-time performance verification

Project Associate

May 2018 – Apr 2019

Indian Institute of Technology Delhi

- Designed hierarchical control systems for quadrotor aerial manipulators with singularity-free algorithms
- Programmed flight control systems and derived complete Lagrangian dynamics for aerial platforms

Education

Ph.D. in Mechanical Engineering

Aug 2019 – May 2023

University of Florida — GPA: 3.97/4.0

Dissertation: “Implicit and Deep Learning-Based Control Methods for Uncertain Nonlinear Systems”

M.S. in Mechanical Engineering

Aug 2019 – Aug 2022

University of Florida — Specialization: Adaptive and Nonlinear Control

B.Tech in Production and Industrial Engineering

Jul 2014 – Aug 2018

Indian Institute of Technology Delhi

Technical Skills

AI & Machine Learning: Deep neural networks, physics-informed neural networks (PINNs), reinforcement learning, online learning, ResNet/LSTM architectures, TensorFlow, PyTorch, concurrent learning algorithms

Control Systems: Lyapunov stability theory, adaptive control, nonlinear control, optimal control, model predictive control (MPC), control barrier functions (CBFs), multi-agent systems, distributed control

Programming: Python, C++, MATLAB/Simulink, ROS, Git, numerical optimization, real-time embedded systems

Robotics & Hardware: Quadrotor UAVs, robotic manipulators, multi-agent platforms, sensor integration, hardware-in-the-loop (HIL) simulation, embedded control systems

Key Achievements & Impact

- **Industry-Ready Innovation:** First to extend Lyapunov adaptive control to deep neural networks of arbitrary depth—enabling real-time AI adaptation with mathematical guarantees for safety-critical systems
- **Funding Success:** Secured \$2M+ from government agencies (ONR, AFOSR, AFRL) demonstrating strong proposal writing and research impact
- **Publications & Recognition:** 20+ papers in premier AI/robotics/control venues with 300+ citations; book contract with Springer on "Lyapunov-Based Deep Learning for Autonomy"
- **Practical Deployment:** Validated algorithms on real robotic hardware including quadrotors, manipulators, and multi-agent testbeds at UF Autonomy Park
- **Open Source Contributions:** Developed widely-used MATLAB/Python packages for adaptive control and neural network controllers
- **Cross-disciplinary Expertise:** Proven ability to bridge theoretical AI/ML with practical engineering applications in aerospace, robotics, and autonomous systems

Selected Publications & Patents

Book: *Lyapunov-Based Deep Learning for Autonomy*, Springer (in development)

Key Papers: 1 book, 8 journal articles, 11+ conference papers in IEEE TAC, Automatica, IEEE T-NN, ACC, CDC

Full Publication List: [Google Scholar Profile](#) 

Awards & Recognition

- **Graduate Student Research Award** (2023) – University of Florida, Department of Mechanical & Aerospace Engineering
- **BOSS Award** (2018) – Best experimental project in Mechanical Engineering, IIT Delhi